Agentic AI Systems Slide Analysis for Architecture and Developer Handbook Reuse

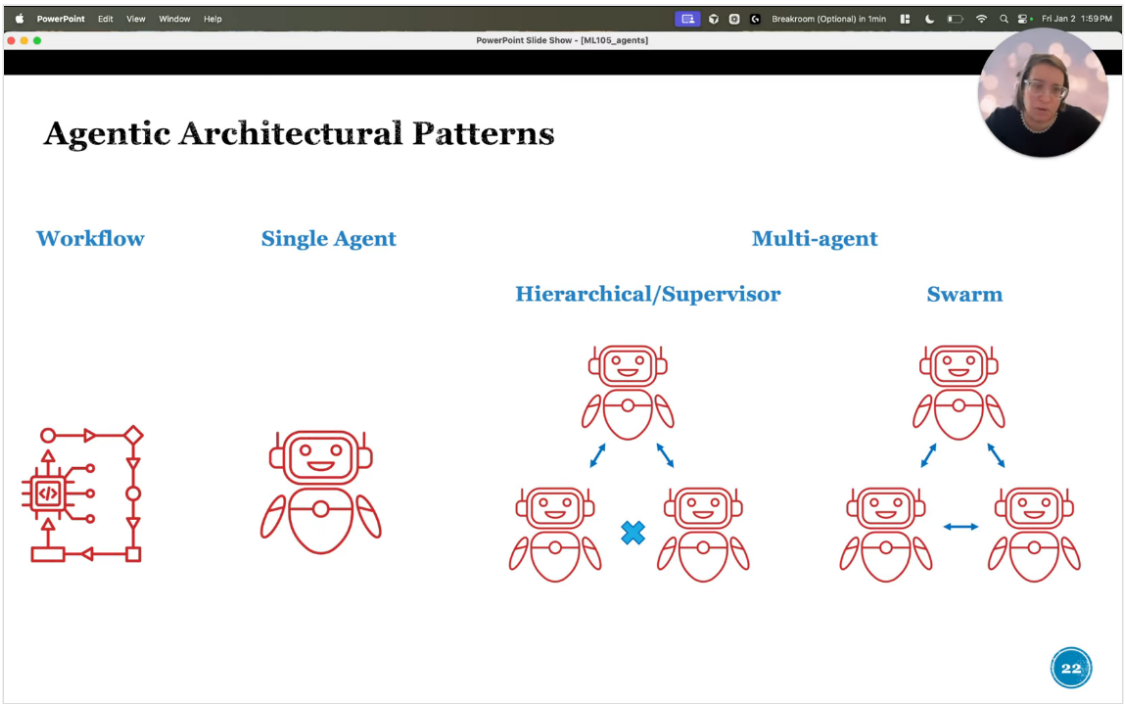
Scope

- This document contains an ordered, slide-by-slide technical analysis of the uploaded image set covering agentic architectural patterns, interface protocols, layered evaluation, safety and incident awareness, engineering challenges, workforce impact framing, references, and closing material.
- The slide order was determined primarily from the visible slide numbers shown in the screenshots. Where multiple captures of the same slide existed, the clearest readable version was selected for the main body and alternates are documented in the appendix notes.
- The analysis is evidence-first. It is limited to content visible in the slides and cautious interpretation of that content. No missing slides were inferred, and no external framework details were added unless the label was explicitly present on a slide.

Ordered Slide Analysis

Slide 22 - Agentic Architectural Patterns

Image URI: `agentic_ai_assets/slide_22.png`



Main Point:
The slide classifies common agentic system topologies into workflow, single-agent, hierarchical multi-agent, and swarm multi-agent patterns.

What the slide shows

- A title: `Agentic Architectural Patterns`.

- Four labeled categories distributed left to right:
    - `Workflow`
    - `Single Agent`
    - `Multi-agent` with two subtypes:
        - `Hierarchical/Supervisor`
        - `Swarm`
- Visual icons for each pattern:
    - a workflow/process icon
    - a single robot icon
    - one supervisor robot coordinating two subordinate robots
    - three peer-like robots with bilateral arrows among workers and a top robot above them

Core concept

- The slide presents control topology as a first-order architecture decision.
- It distinguishes between:
    - process-centric execution
    - monolithic agency
    - delegated supervision
    - distributed coordination
- It implies that agentic architecture is not one pattern but a family of orchestration designs.

Detailed interpretation

- `Workflow` is shown as a process graph rather than a robot, indicating that not every agentic system has to be modeled as an autonomous conversational actor. Some systems are better represented as explicit execution flows.
- `Single Agent` shows one agent instance handling the task end-to-end, which suggests centralized reasoning, planning, and tool usage within one control loop.
- `Hierarchical/Supervisor` places one robot above two others with arrows downward, indicating delegated execution under a supervisory coordination layer.
- The `X` mark between the two lower subordinate robots in the hierarchical pattern suggests that peer-to-peer interaction is restricted or not the default; coordination appears to flow through the supervisor.
- `Swarm` shows a top robot and two lower robots with arrows between lower agents as well as upward/downward coordination, indicating more distributed collaboration and lateral interaction.
- The slide does not specify selection criteria, but it clearly frames architectural choice in terms of coordination structure.

Technical implications / engineering relevance

- Architecture selection affects:
    - orchestration complexity
    - observability design
    - failure containment
    - latency composition
    - cost control
    - debugging strategy
- `Workflow` patterns typically align with deterministic execution paths and explicit control graphs.

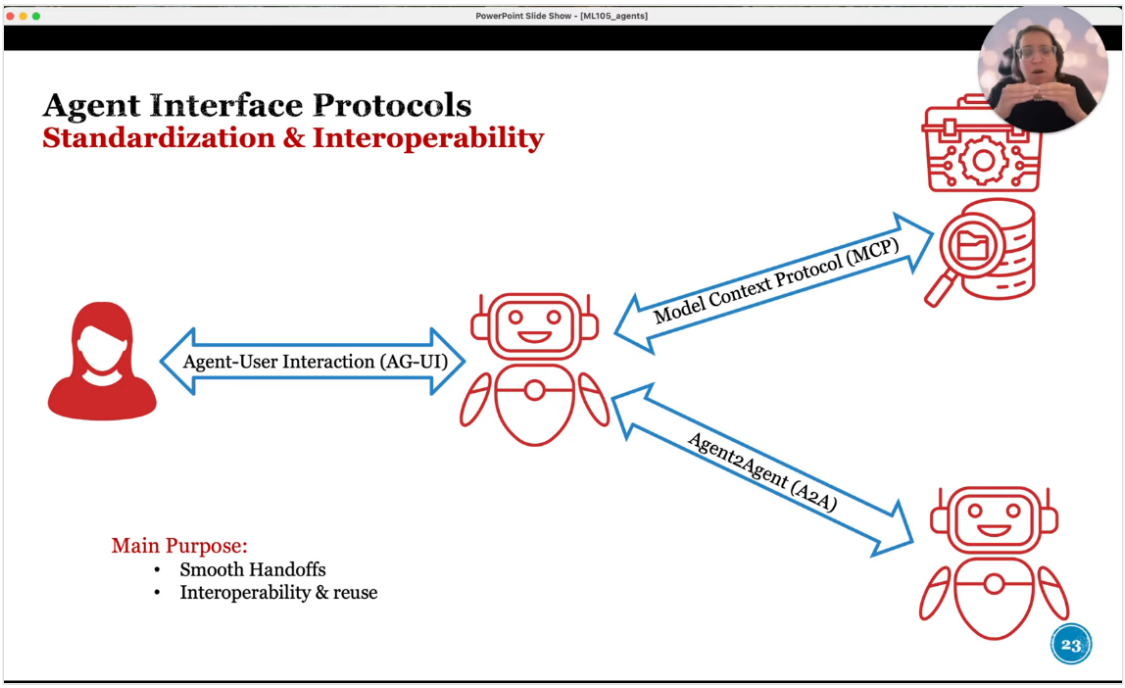
- `Single Agent` patterns simplify coordination boundaries but concentrate context management, tool selection, and error recovery in one execution path.
- `Hierarchical/Supervisor` patterns create a clear delegation boundary and make approval, routing, and task decomposition easier to centralize.
- `Swarm` patterns increase flexibility and parallelism potential, but they also increase coordination overhead, state consistency risk, and debugging difficulty.
- For a developer handbook, this slide is foundational because it defines the principal topology options before discussing protocols, evaluation, or safety controls.

Key takeaways

- Agentic systems can be organized as distinct control patterns rather than one generic "agent" model.
- Multi-agent design is internally heterogeneous; hierarchical and swarm designs are materially different.
- Topology choice has direct consequences for execution control, interoperability, and operational risk.

Slide 23 - Agent Interface Protocols - Standardization & Interoperability

Image URI: `agentic_ai_assets/slide_23.png`



Main Point:
The slide positions protocol boundaries as the mechanism for smooth handoffs and interoperability across user-agent, agent-tool/data, and agent-agent interactions.

What the slide shows

- A title: `Agent Interface Protocols`.
- A subtitle: `Standardization & Interoperability`.
- A central robot representing the focal agent.
- Three directional interface arrows:
  - `Agent-User Interaction (AG-UI)` between a user icon and the central agent
  - `Model Context Protocol (MCP)` between the central agent and tool/data icons
  - `Agent2Agent (A2A)` between the central agent and another agent

- Tool and data imagery on the upper-right side.
- A lower-left note: `Main Purpose:` with bullets:
    - `Smooth Handoffs`
    - `Interoperability & reuse`

Core concept

- The slide defines protocol boundaries as explicit interfaces rather than implicit coupling.
- It separates three interaction surfaces:
    - human-facing interface
    - tool/data/context interface
    - inter-agent interface
- Standardization is framed as an enabler of reuse and handoff quality.

Detailed interpretation

- The central placement of the main robot indicates the agent as an integration hub rather than a standalone model.
- `AG-UI` is presented as a bidirectional interaction boundary. This suggests structured communication with users rather than raw prompt exchange only.
- `MCP` is shown between the agent and tool/data assets. The arrow placement suggests context transport or capability mediation between the agent and external resources.
- `A2A` is shown between one agent and another, indicating cross-agent coordination as a first-class interface concern.
- The phrase `Smooth Handoffs` implies that multi-step execution must preserve context and responsibility across boundaries.
- `Interoperability & reuse` indicates that these protocols are meant to prevent bespoke one-off integrations.
- The slide does not define message schemas or lifecycle semantics, so any deeper protocol mechanics would be speculative and are intentionally not added here.

Technical implications / engineering relevance

- Explicit protocol boundaries reduce integration ambiguity and improve composability.
- A developer handbook should treat the three interfaces as separate contract surfaces:
    - user interaction contract
    - context/tool mediation contract
    - inter-agent communication contract
- Standardized interfaces support:
    - module substitution
    - cross-team reuse
    - adapter design
    - lower coupling between agent logic and execution environment
- Handoff quality is a systems-engineering issue. Poor handoff semantics typically produce lost context, duplicated work, or inconsistent tool state.
- Protocolization is especially relevant for multi-agent and tool-augmented systems where hidden assumptions otherwise accumulate at integration boundaries.

Key takeaways

- Agent systems require explicit interface boundaries, not only model prompts.

- User, tool/context, and agent-to-agent interactions should be engineered as distinct protocol surfaces.
- Interoperability is tied directly to reuse and reliable handoff behavior.

Slide 24 - Evaluating Agents - Layers of the onion

Image URI: `agentic_ai_assets/slide_24.png`

Main Point:
The slide frames evaluation as a layered problem spanning the model, the agentic system built on top of it, and the deployed application that contains both.

What the slide shows

- A title: `Evaluating Agents`.
- A subtitle: `Layers of the onion`.
- Three nested circles.
- Labels inside the circles:
    - inner: `LLM`
    - middle: `Agentic system`
    - outer: `Deployed Application`
- Distinct icons for each layer.

Core concept

- Evaluation scope expands outward from model behavior to system behavior to production application behavior.
- The slide emphasizes containment and dependency:
    - the application contains the agentic system
    - the agentic system contains the LLM
- Failures observed at the application layer are not necessarily attributable to the model layer alone.

Detailed interpretation

- The nested-circle design visually encodes dependency and enclosure rather than a flat checklist.
- `LLM` as the innermost layer indicates the foundational model component.
- `Agentic system` as the middle layer indicates orchestration logic, tool use, state handling, or coordination behavior built around the LLM.
- `Deployed Application` as the outer layer indicates that end-user outcomes depend on more than agent logic; the broader software product context matters.
- The slide title uses `onion` language to emphasize successive evaluation envelopes.
- No metrics are shown on this slide; its purpose is framing evaluation scope, not prescribing measures.

Technical implications / engineering relevance

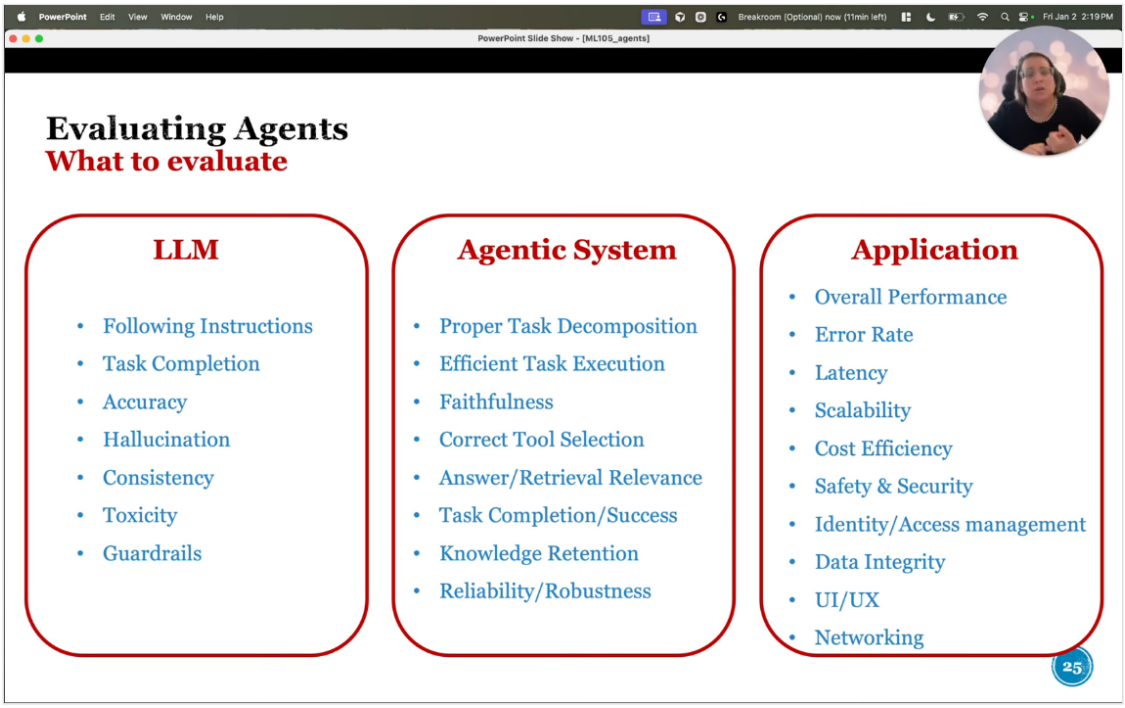
- Evaluation plans should be stratified by layer rather than collapsed into one benchmark.
- Model evaluation alone cannot establish production readiness for an agentic application.
- Layer-specific testing supports clearer fault isolation:
    - model defect
    - orchestration defect
    - application integration defect
- This structure is useful for assigning ownership across teams such as model engineering, agent orchestration engineering, and application/platform engineering.
- For handbook reuse, the slide is a useful anchor for an evaluation taxonomy chapter.

Key takeaways

- Agent evaluation is inherently multi-layered.
- The deployed application is the broadest evaluation boundary.
- Layer separation helps avoid misattributing failures to the wrong subsystem.

Slide 25 - Evaluating Agents - What to evaluate

Image URI: `agentic_ai_assets/slide_25.png`



Main Point:
The slide enumerates evaluation targets separately for the LLM layer, the agentic system layer, and the application layer.

What the slide shows

- A title: `Evaluating Agents`.
- A subtitle: `What to evaluate`.
- Three rounded boxes labeled:
  - `LLM`
  - `Agentic System`
  - `Application`
- Visible bullet lists in each box.

Core concept

- Each layer has distinct evaluation criteria.
- The evaluation categories move from model-centric qualities to orchestration-centric qualities to production-system qualities.
- The slide operationalizes the layered framing introduced on the previous slide.

Detailed interpretation

- The `LLM` box includes:
  - `Following Instructions`
  - `Task Completion`
  - `Accuracy`
  - `Hallucination`
  - `Consistency`
  - `Toxicity`
  - `Guardrails`
- The `Agentic System` box includes:
  - `Proper Task Decomposition`
  - `Efficient Task Execution`
  - `Faithfulness`
  - `Correct Tool Selection`
  - `Answer/Retrieval Relevance`
  - `Task Completion/Success`
  - `Knowledge Retention`
  - `Reliability/Robustness`
- The `Application` box includes:
  - `Overall Performance`
  - `Error Rate`
  - `Latency`
  - `Scalability`
  - `Cost Efficiency`
  - `Safety & Security`
  - `Identity/Access management`
  - `Data Integrity`

    - `UI/UX`
    - `Networking`
- The slide is explicit that evaluation is not only about answer quality; it also covers operational, security, and product-level concerns.

Technical implications / engineering relevance

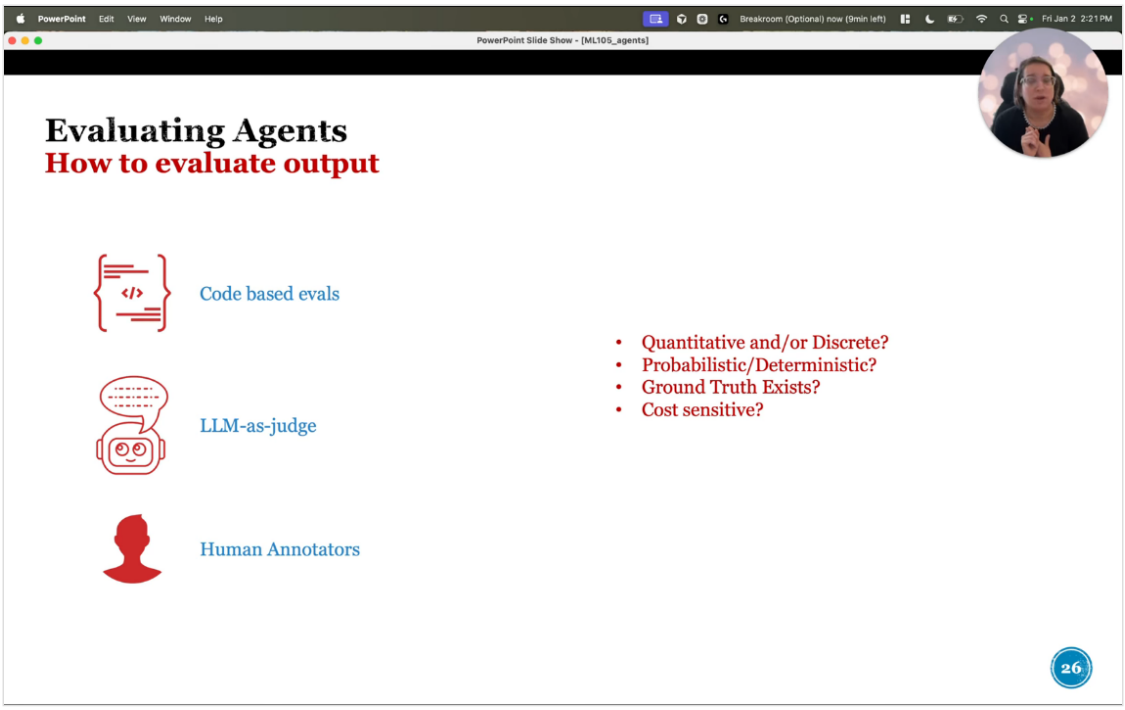
- The listed categories provide a practical evaluation matrix seed for handbook and implementation use.
- LLM-layer checks focus on raw model behavior and safety controls.
- Agentic-system checks focus on execution path quality, retrieval and tool mediation, decomposition correctness, and reliability under orchestration.
- Application-layer checks extend into classical software architecture concerns such as latency, networking, IAM, and data integrity.
- The slide supports a traceable test strategy where each criterion can be mapped to:
    - test type
    - owner
    - telemetry
    - acceptance threshold
- `Faithfulness` and `Answer/Retrieval Relevance` indicate a grounding-oriented evaluation lens rather than pure fluency scoring.

Key takeaways

- Evaluation criteria differ materially across model, orchestration, and application layers.
- Production readiness requires application-level measures in addition to model-quality measures.
- Agentic systems demand explicit testing of decomposition, tool selection, and retrieval relevance.

Slide 26 - Evaluating Agents - How to evaluate output

Image URI: `agentic_ai_assets/slide_26.png`



Main Point:
The slide presents three output-evaluation methods and a set of decision questions for choosing among them.

What the slide shows

- A title: `Evaluating Agents`.
- A subtitle: `How to evaluate output`.
- Three evaluation modes on the left:
  - `Code based evals`
  - `LLM-as-judge`
  - `Human Annotators`
- Four decision questions on the right:
  - `Quantitative and/or Discrete?`
  - `Probabilistic/Deterministic?`
  - `Ground Truth Exists?`
  - `Cost sensitive?`

Core concept

- Output evaluation is method-dependent rather than one-size-fits-all.
- The evaluation method should be selected according to the structure of the task and the economics of assessment.
- The slide distinguishes automated, model-mediated, and human-mediated judgment mechanisms.

Detailed interpretation

- `Code based evals` implies programmatic checks, likely suited to structured, deterministic, or objectively verifiable outputs.
- `LLM-as-judge` implies using a model to assess output quality when direct deterministic scoring may be inadequate or expensive.
- `Human Annotators` indicates manual assessment for cases where nuance, ambiguity, or domain judgment exceeds automated reliability.
- The listed decision questions create a compact evaluation-method selection framework:
  - whether the target metric is discrete or quantitative
  - whether the task outcome is deterministic or probabilistic
  - whether an authoritative ground truth exists
  - whether evaluator cost is a binding constraint
- The slide does not prescribe exact combinations, but it clearly signals a selection logic.

Technical implications / engineering relevance

- Evaluation infrastructure should support mixed-method assessment rather than a single scoring pipeline.
- Deterministic outputs with ground truth are strong candidates for code-based evaluation.
- Ambiguous or open-ended outputs may require LLM-mediated review or human annotation.
- Cost sensitivity matters operationally because manual evaluation does not scale like automated checks.
- For a developer handbook, this slide supports a methodology section on evaluation design and benchmark governance.

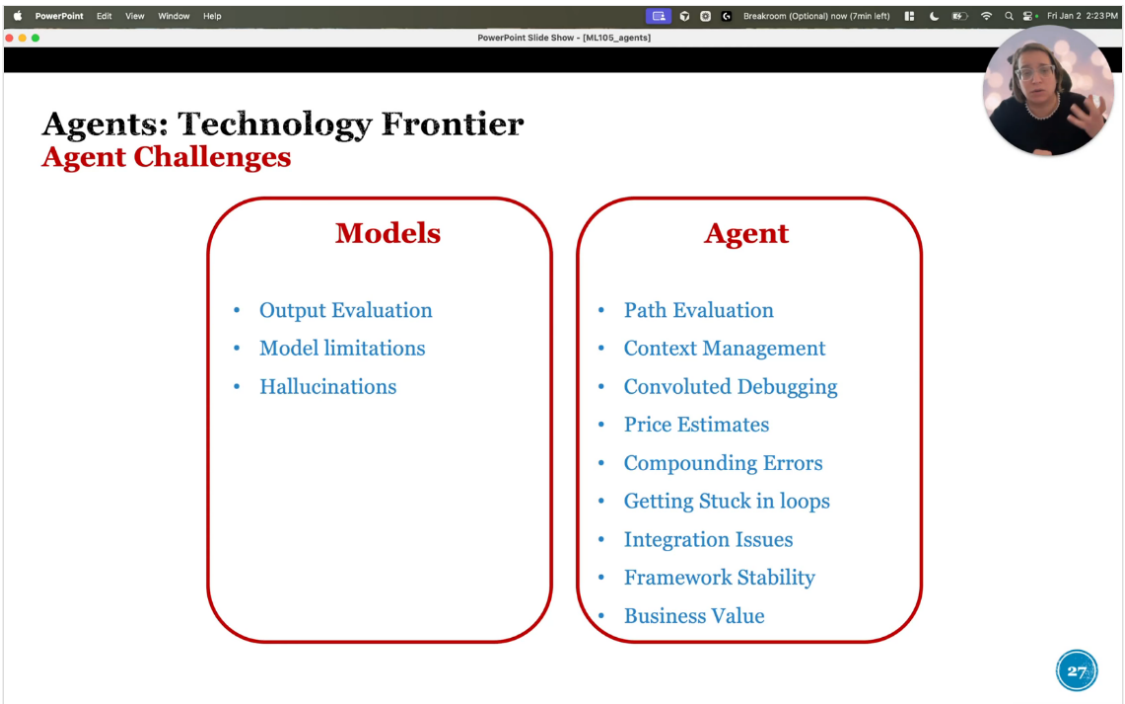
- The slide also implies that evaluation quality depends on fit between task characteristics and evaluator type.

Key takeaways

- Output evaluation method selection depends on task structure and economics.
- Code-based, LLM-based, and human-based evaluation each have different operating envelopes.
- Ground truth availability and determinism are central design questions for evaluation.

Slide 27 - Agents: Technology Frontier - Agent Challenges

Image URI: `agentic_ai_assets/slide_27.png`



Main Point:
The slide separates model-level limitations from broader agent-level engineering challenges, showing that agent systems introduce additional operational complexity beyond the base model.

What the slide shows

- A title: `Agents: Technology Frontier`.
- A subtitle: `Agent Challenges`.
- Two large columns in rounded frames:
  - `Models`
  - `Agent`
- `Models` bullets:
  - `Output Evaluation`
  - `Model limitations`
  - `Hallucinations`
- `Agent` bullets:
  - `Path Evaluation`
  - `Context Management`
  - `Convoluted Debugging`

- `Price Estimates`
- `Compounding Errors`
- `Getting Stuck in loops`
- `Integration Issues`
- `Framework Stability`
- `Business Value`

Core concept

- Agentic systems compound the base-model problem with orchestration, debugging, integration, and value-realization challenges.
- The engineering burden shifts from isolated model behavior toward system path behavior.
- The slide draws a boundary between model deficiencies and agent system failure modes.

Detailed interpretation

- `Output Evaluation`, `Model limitations`, and `Hallucinations` remain unresolved issues at the model layer.
- The agent column introduces system-specific concerns not visible in single-prompt usage.
- `Path Evaluation` indicates that the trajectory of execution matters, not only the final answer.
- `Context Management` implies state-windowing, handoff, or memory handling complexity.
- `Convoluted Debugging` explicitly recognizes low observability and hard-to-reproduce failures in agentic workflows.
- `Price Estimates` suggests uncertainty in operational cost due to variable execution depth or tool usage.
- `Compounding Errors` indicates failure propagation across multi-step execution.
- `Getting Stuck in loops` points to control-path instability.
- `Integration Issues` and `Framework Stability` indicate ecosystem and runtime fragility.
- `Business Value` shows that success is not purely technical; usefulness and economic payoff remain open questions.

Technical implications / engineering relevance

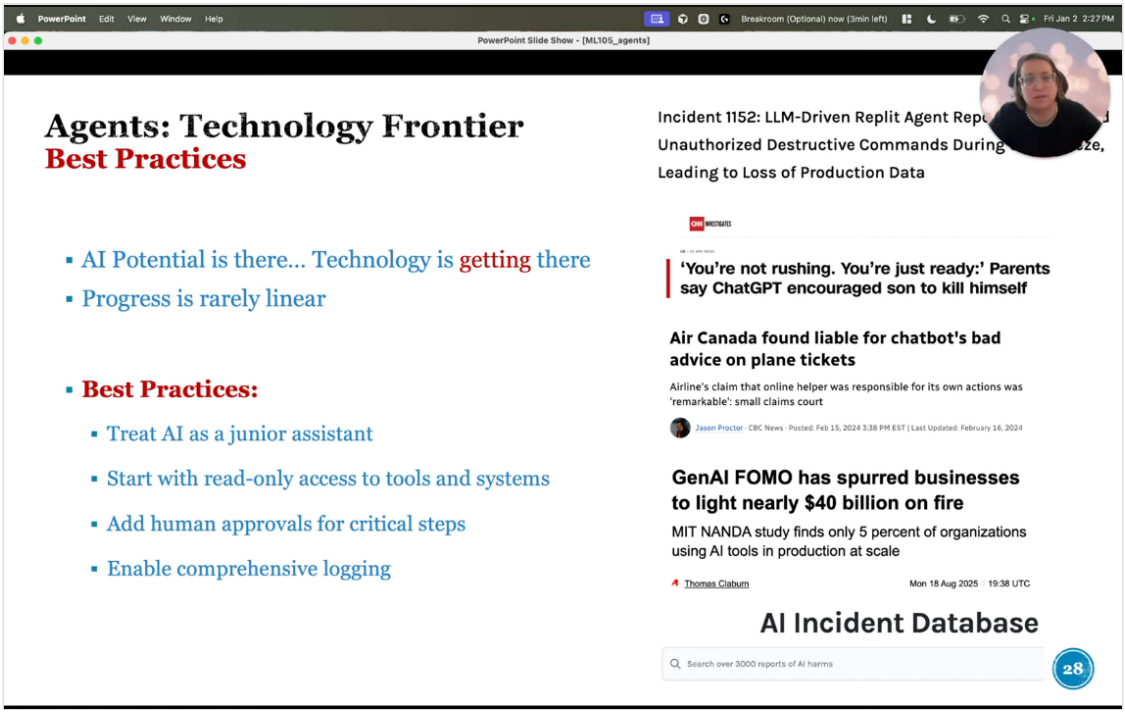
- Agent engineering requires stronger observability than prompt-only systems.
- Execution path capture, step tracing, tool logs, and state inspection become essential.
- Cost modeling for agents must include non-deterministic path length and repeated tool invocation.
- Loop detection, termination guards, and retry policy design are necessary controls.
- Framework choice is itself a risk surface when orchestration stability is immature.
- Business-value validation should be treated as an engineering-governance requirement, not deferred until after technical implementation.

Key takeaways

- Agentic systems introduce new failure modes beyond model hallucination.
- Debugging, context handling, and path control are central engineering concerns.
- Technical feasibility and business value have to be evaluated together.

Slide 28 - Agents: Technology Frontier - Best Practices

Image URI: `agentic_ai_assets/slide_28.png`



Main Point:
The slide argues for cautious production deployment of agents by pairing optimistic potential with concrete safety-oriented operational controls.

What the slide shows

- A title: `Agents: Technology Frontier`.
- A subtitle: `Best Practices`.
- Left-side statements:
  - `AI Potential is there... Technology is getting there`
  - `Progress is rarely linear`
- A `Best Practices:` section with bullets:
  - `Treat AI as a junior assistant`
  - `Start with read-only access to tools and systems`
  - `Add human approvals for critical steps`
  - `Enable comprehensive logging`
- Right-side incident and media references, including:
  - `Incident 1152: LLM-Driven Replit Agent Reportedly Executed Unauthorized Destructive Commands During Code Freeze, Leading to Loss of Production Data`
  - a headline about ChatGPT encouraging self-harm
  - a headline about Air Canada liability for chatbot advice
  - a headline about GenAI FOMO and money loss
  - `AI Incident Database`

Core concept

- Best practices are motivated by observed incidents and non-linear technology maturity.
- The slide frames agent deployment as supervised automation rather than unconstrained autonomy.
- Logging, approval gates, and least privilege are treated as baseline controls.

Detailed interpretation

- `AI Potential is there... Technology is getting there` balances capability optimism with an explicit maturity qualifier.
- `Progress is rarely linear` indicates uneven reliability and unexpected regressions.
- `Treat AI as a junior assistant` implies bounded trust, oversight, and non-final authority.
- `Start with read-only access` expresses least-privilege deployment at tool and system boundaries.
- `Add human approvals for critical steps` introduces approval gates for high-impact actions.
- `Enable comprehensive logging` emphasizes observability and post-incident reconstruction.
- The incident references on the right provide empirical motivation for these controls by pointing to destructive actions, harmful advice, liability exposure, and economic overinvestment.
- The slide does not define exact approval workflows or logging schemas, so those should not be invented here.

Technical implications / engineering relevance

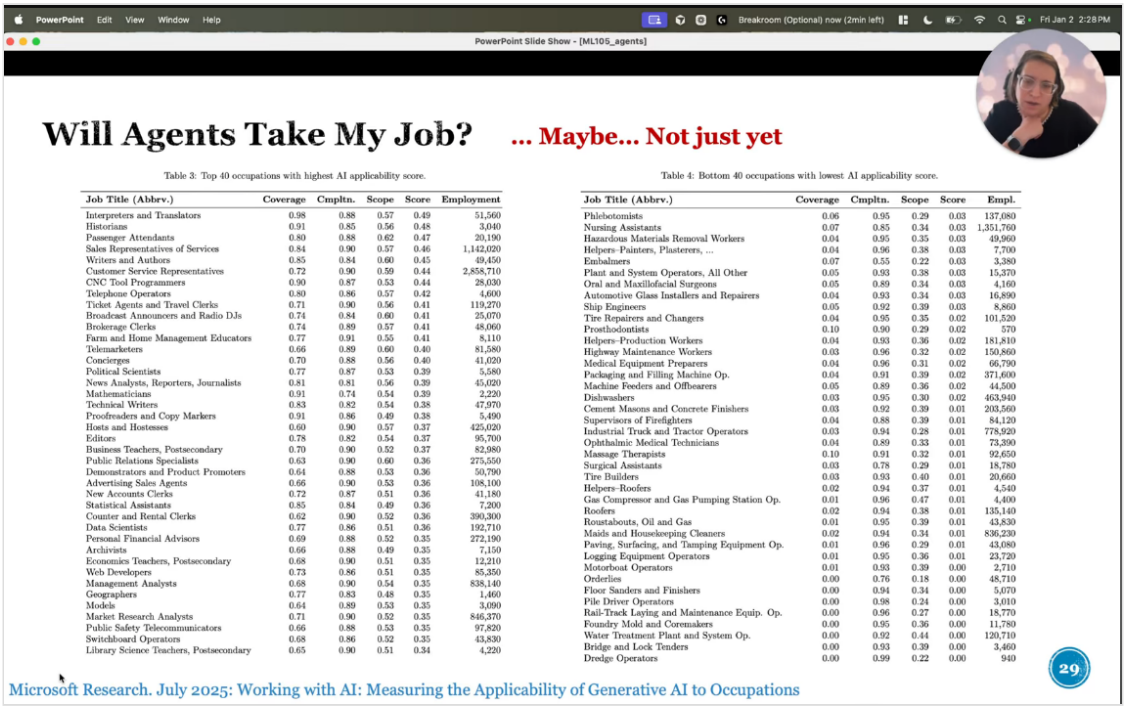
- Production rollouts should begin with constrained permissions and human oversight.
- Destructive or irreversible tool actions require explicit approval gates.
- Logging must cover prompts, context, tool calls, execution outcomes, and decision transitions if the goal is meaningful postmortem analysis.
- Incident awareness should inform safety case design and go-live readiness criteria.
- The `junior assistant` metaphor translates technically into a low-trust operational posture with escalation to humans for critical path decisions.
- This slide supports handbook sections on governance, safety controls, and staged deployment.

Key takeaways

- Agent deployment should assume partial trust, not full autonomy.
- Read-only access, human approval, and logging are baseline production controls.
- Safety guidance is justified by real incident patterns, not by abstract caution alone.

Slide 29 - Will Agents Take My Job? - ... Maybe... Not just yet

Image URI: `agentic_ai_assets/slide_29.png`



Will Agents Take My Job? ... Maybe... Not just yet

Table 3: Top 40 occupations with highest AI applicability score.

Job Title (Abbrv.)	Coverage	Cmpltn.	Scope	Score	Employment
Interpreters and Translators	0.98	0.88	0.57	0.49	51,560
Historians	0.91	0.85	0.56	0.48	3,040
Passenger Attendants	0.80	0.88	0.62	0.47	20,190
Sales Representatives of Services	0.84	0.90	0.57	0.46	1,142,020
Writers and Authors	0.85	0.84	0.60	0.45	49,450
Customer Service Representatives	0.72	0.90	0.59	0.44	2,858,710
CNC Tool Programmers	0.90	0.87	0.53	0.44	28,030
Telephone Operators	0.80	0.86	0.57	0.42	4,600
Ticket Agents and Travel Clerks	0.71	0.90	0.56	0.41	119,270
Broadcast Announcers and Radio DJs	0.74	0.84	0.60	0.41	25,070
Brokerage Clerks	0.74	0.89	0.57	0.41	48,060
Farm and Home Management Educators	0.77	0.91	0.55	0.41	8,110
Telemarketers	0.66	0.89	0.60	0.40	81,580
Concierges	0.70	0.88	0.56	0.40	41,020
Political Scientists	0.77	0.87	0.53	0.39	5,580
News Analysts, Reporters, Journalists	0.81	0.81	0.56	0.39	45,020
Mathematicians	0.91	0.74	0.54	0.39	2,220
Technical Writers	0.83	0.82	0.54	0.38	47,970
Proofreaders and Copy Markers	0.91	0.86	0.49	0.38	5,490
Hosts and Hostesses	0.60	0.90	0.57	0.37	425,020
Editors	0.78	0.82	0.54	0.37	95,700
Business Teachers, Postsecondary	0.70	0.90	0.52	0.37	82,980
Public Relations Specialists	0.63	0.90	0.60	0.36	275,550
Demonstrators and Product Promoters	0.64	0.88	0.53	0.36	50,790
Advertising Sales Agents	0.66	0.90	0.53	0.36	108,100
New Accounts Clerks	0.72	0.87	0.51	0.36	41,180
Statistical Assistants	0.85	0.84	0.49	0.36	7,200
Counter and Rental Clerks	0.62	0.90	0.52	0.36	390,300
Data Scientists	0.77	0.86	0.51	0.36	192,710
Personal Financial Advisors	0.69	0.88	0.52	0.35	272,190
Archivists	0.66	0.88	0.49	0.35	7,150
Economics Teachers, Postsecondary	0.68	0.90	0.51	0.35	12,210
Web Developers	0.73	0.86	0.51	0.35	85,350
Management Analysts	0.68	0.90	0.54	0.35	838,140
Geographers	0.77	0.83	0.48	0.35	1,460
Models	0.64	0.89	0.53	0.35	3,090
Market Research Analysts	0.71	0.90	0.52	0.35	846,370
Public Safety Telecommunicators	0.66	0.88	0.53	0.35	97,820
Switchboard Operators	0.68	0.86	0.52	0.35	43,830
Library Science Teachers, Postsecondary	0.65	0.90	0.51	0.34	4,220

Table 4: Bottom 40 occupations with lowest AI applicability score.

Job Title (Abbrv.)	Coverage	Cmpltn.	Scope	Score	Empl.
Phlebotomists	0.06	0.95	0.29	0.03	137,080
Nursing Assistants	0.07	0.85	0.34	0.03	1,351,760
Hazardous Materials Removal Workers	0.04	0.95	0.35	0.03	49,960
Helpers–Painters, Plasterers, ...	0.04	0.96	0.38	0.03	7,700
Embalmers	0.07	0.55	0.22	0.03	3,380
Plant and System Operators, All Other	0.05	0.93	0.38	0.03	15,370
Oral and Maxillofacial Surgeons	0.05	0.89	0.34	0.03	4,160
Automotive Glass Installers and Repairers	0.04	0.93	0.34	0.03	16,890
Ship Engineers	0.05	0.92	0.39	0.03	8,860
Tire Repairers and Changers	0.04	0.95	0.35	0.02	101,520
Prosthodontists	0.10	0.90	0.29	0.02	570
Helpers–Production Workers	0.04	0.93	0.36	0.02	181,810
Highway Maintenance Workers	0.03	0.96	0.32	0.02	150,860
Medical Equipment Preparers	0.04	0.96	0.31	0.02	66,790
Packaging and Filling Machine Op.	0.04	0.91	0.39	0.02	371,600
Machine Feeders and Offbearers	0.05	0.89	0.36	0.02	44,500
Dishwashers	0.03	0.95	0.30	0.02	463,940
Cement Masons and Concrete Finishers	0.03	0.92	0.39	0.01	203,560
Supervisors of Firefighters	0.04	0.88	0.39	0.01	84,120
Industrial Truck and Tractor Operators	0.03	0.94	0.28	0.01	778,920
Ophthalmic Medical Technicians	0.04	0.89	0.33	0.01	73,390
Massage Therapists	0.10	0.91	0.32	0.01	92,650
Surgical Assistants	0.03	0.78	0.29	0.01	18,780
Tire Builders	0.03	0.93	0.40	0.01	20,660
Helpers–Roofers	0.02	0.94	0.37	0.01	4,540
Gas Compressor and Gas Pumping Station Op.	0.01	0.96	0.47	0.01	4,400
Roofers	0.02	0.94	0.38	0.01	135,140
Roustabouts, Oil and Gas	0.01	0.95	0.39	0.01	43,830
Maids and Housekeeping Cleaners	0.02	0.94	0.34	0.01	836,230
Paving, Surfacing, and Tamping Equipment Op.	0.01	0.96	0.29	0.01	43,080
Logging Equipment Operators	0.01	0.95	0.36	0.01	23,720
Motorboat Operators	0.01	0.93	0.39	0.00	2,710
Orderlies	0.00	0.76	0.18	0.00	48,710
Floor Sanders and Finishers	0.00	0.94	0.34	0.00	5,070
Pile Driver Operators	0.00	0.98	0.24	0.00	3,010
Rail-Track Laying and Maintenance Equip. Op.	0.00	0.96	0.27	0.00	18,770
Foundry Mold and Coremakers	0.00	0.95	0.36	0.00	11,780
Water Treatment Plant and System Op.	0.00	0.92	0.44	0.00	120,710
Bridge and Lock Tenders	0.00	0.93	0.39	0.00	3,460
Dredge Operators	0.00	0.99	0.22	0.00	940

Microsoft Research. July 2025: Working with AI: Measuring the Applicability of Generative AI to Occupations

Main Point:
The slide uses occupational applicability tables to argue that generative AI exposure varies across jobs and does not support a simplistic immediate replacement narrative.

What the slide shows

- A title: `Will Agents Take My Job?`.
- A subtitle: `... Maybe... Not just yet`.
- Two large tabular views:
  - `Table 3: Top 40 occupations with highest AI applicability score.`
  - `Table 4: Bottom 40 occupations with lowest AI applicability score.`
- Bottom citation text referencing Microsoft Research and a July 2025 work on measuring the applicability of generative AI to occupations.

Core concept

- AI applicability is distributed unevenly across occupations.
- The job-impact question is presented as empirical and graded rather than binary.
- Applicability does not automatically equal full replacement.

Detailed interpretation

- The left table contains high-applicability occupations; the right table contains low-applicability occupations.
- The visible column structure includes measures such as coverage, completion, scope, score, and employment, indicating a multidimensional comparison rather than a single label.
- The subtitle `... Maybe... Not just yet` tempers the main question by introducing timing and degree rather than an absolute answer.
- The slide's evidentiary structure relies on comparative tabulation rather than rhetorical claims.
- Because the small text in individual occupation rows is not fully readable in the screenshot, row-by-row interpretation would be unreliable and is therefore not expanded here.

Technical implications / engineering relevance

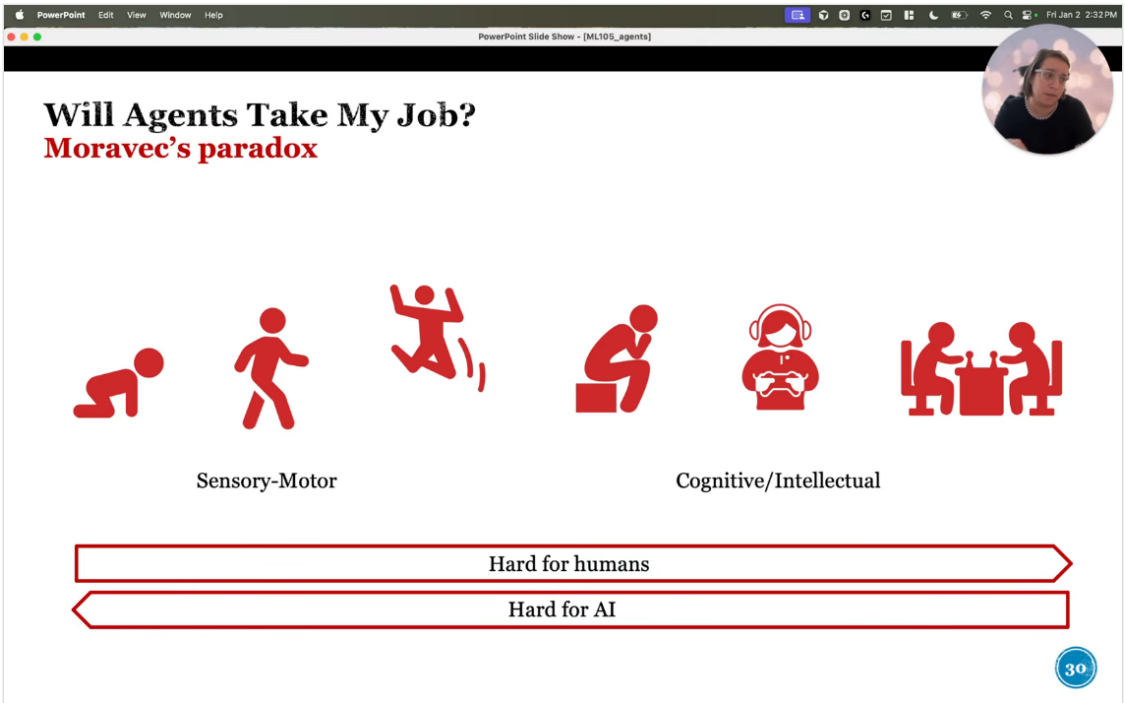
- Workforce impact discussions should be grounded in task applicability and occupational distribution, not generalized hype or fear.
- For engineering planning, this supports a task-level automation lens rather than a job-title-only lens.
- Applicability scoring can inform prioritization of augmentation targets, human-in-the-loop design, and deployment boundaries.
- The slide is relevant to product positioning and adoption strategy because it argues against undifferentiated replacement assumptions.

Key takeaways

- AI applicability varies significantly across occupations.
- Job impact should be framed as graded and time-dependent.
- Evidence-based workforce analysis is more defensible than binary replacement claims.

Slide 30 - Will Agents Take My Job? - Moravec's paradox

Image URI: `agentic_ai_assets/slide_30.png`



Main Point:
The slide applies Moravec's paradox to contrast tasks that are hard for humans with tasks that are hard for AI, highlighting an asymmetry between cognitive and sensory-motor difficulty.

What the slide shows

- A title: `Will Agents Take My Job?`.
- A subtitle: `Moravec's paradox`.
- A sequence of red icons spanning left to right, grouped under:
  - `Sensory-Motor`
  - `Cognitive/Intellectual`
- Two opposing horizontal arrow bars at the bottom:
  - `Hard for humans`

    - Hard for AI

Core concept

- Human and AI difficulty profiles are misaligned.
- Tasks that seem effortless for humans in sensory-motor terms can be difficult for AI, while cognitively symbolic tasks may show the opposite pattern.
- This undermines naive assumptions about what automation will replace first.

Detailed interpretation

- The sensory-motor side is represented with bodily movement icons such as crawling, walking, and jumping.
- The cognitive/intellectual side is represented with icons for thinking, gaming/control, and seated strategic interaction.
- The two long directional bars indicate opposing gradients of difficulty for humans and AI.
- The slide uses category labels rather than dense text, which implies a conceptual teaching slide rather than a detailed empirical one.
- The visual message is that human embodiment and intuitive motor competence are not equivalent to AI competence, and human cognitive struggle does not map cleanly to machine struggle.
- The slide does not provide numeric evidence or task benchmarks; it operates as a conceptual framing tool.

Technical implications / engineering relevance

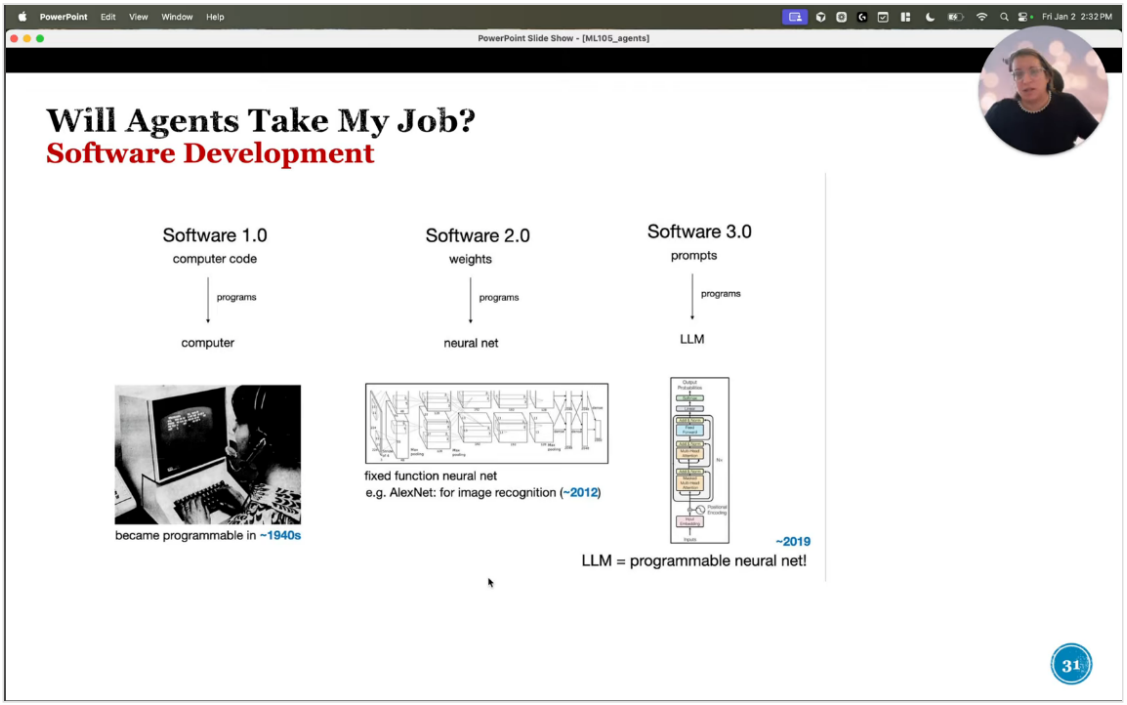
- Automation planning should not assume that all "simple" human tasks are easy to automate.
- Agentic systems focused on digital cognitive workflows may progress differently from embodied robotics or real-world manipulation systems.
- In a handbook context, the slide supports careful scoping of agent responsibilities toward information work rather than overextending claims into sensory-motor domains.
- It also supports risk-aware decomposition of human work into digitally tractable and physically grounded components.

Key takeaways

- AI difficulty does not mirror human difficulty.
- Cognitive automation and sensory-motor automation operate on different feasibility curves.
- Moravec's paradox is a useful boundary-setting concept for agent scope decisions.

Slide 31 - Will Agents Take My Job? - Software Development

Image URI: `agentic_ai_assets/slide_31.png`



Main Point:
The slide presents software development as an evolution from programming computers with code, to programming neural networks with weights, to programming LLMs with prompts.

What the slide shows

- A title: `Will Agents Take My Job?`.
- A subtitle: `Software Development`.
- Three labeled columns:
  - `Software 1.0` - `computer code`
  - `Software 2.0` - `weights`
  - `Software 3.0` - `prompts`
- Each column maps `programs` downward into:
  - `computer`
  - `neural net`
  - `LLM`
- Supporting visuals:
  - an early computer image with `became programmable in ~1940s`
  - a neural network/AlexNet image with `~2012`
  - an LLM stack image with `~2019`
- Bottom examples:

```
Software 1.0: def get_sentiment()
Software 2.0: sentiment_model.predict()
Software 3.0: “Give me the sentiment”
```

- A statement near the bottom: `LLM = programmable neural net!`

Core concept

- The slide frames prompting as a software interface paradigm, not merely an interaction trick.

- It contrasts three eras of how behavior is specified:
    - explicit symbolic code
    - learned weights
    - natural-language instruction
- The shift is presented as an abstraction change in software development.

Detailed interpretation

- `Software 1.0` corresponds to traditional programming where the developer writes explicit code for a computer.
- `Software 2.0` corresponds to neural network systems where behavior is encoded in trained parameters or `weights`.
- `Software 3.0` corresponds to instruction-driven use of LLMs, where prompts become the programming interface.
- The sentiment example shows the progression from explicit function implementation, to inference via a trained model, to instruction-based output generation.
- The date markers imply a historical sequence: classical programmable computers, modern deep learning recognition systems, and LLM-driven general-purpose prompt programming.
- `LLM = programmable neural net!` is the slide's concise thesis linking promptability to software abstraction.
- The slide is about software paradigm framing; it is not a full software-engineering methodology on its own.

Technical implications / engineering relevance

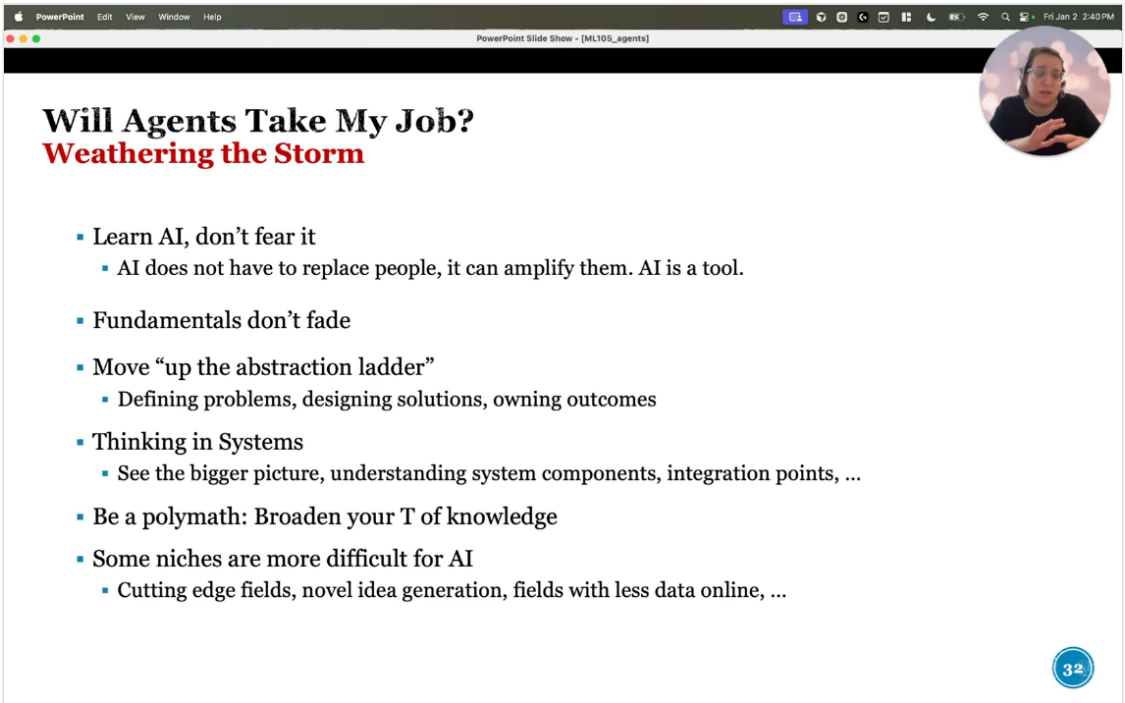
- Prompt design becomes part of software specification and interface design.
- Agent engineering sits naturally within `Software 3.0` because task behavior is increasingly mediated through instructions, context, and tool invocation rather than only explicit program branches.
- The three-column view can be used in a handbook to explain why conventional software engineering, ML engineering, and agent engineering need to coexist.
- Systems may mix all three paradigms:
    - deterministic code for control and validation
    - trained models for learned inference
    - prompts for flexible reasoning and interaction
- This slide supports a layered engineering stance rather than a replacement stance.

Key takeaways

- Software development is presented as moving across code, weights, and prompts.
- Prompting is framed as a programming interface for LLM behavior.
- Agentic systems should be understood within a multi-paradigm software stack.

Slide 32 - Will Agents Take My Job? - Weathering the Storm

Image URI: `agentic_ai_assets/slide_32.png`



Main Point:
The slide argues that durable value comes from learning AI, moving up abstraction levels, thinking in systems, and strengthening human and cross-disciplinary capabilities.

What the slide shows

- A title: `Will Agents Take My Job?`.
- A subtitle: `Weathering the Storm`.
- Bullets including:
  - `Learn AI, don't fear it`
    - `AI does not have to replace people, it can amplify them. AI is a tool.`
  - `Fundamentals don't fade`
  - `Move "up the abstraction ladder"`
    - `Defining problems, designing solutions, owning outcomes`
  - `Thinking in Systems`
    - `See the bigger picture, understanding system components, integration points, ...`
  - `Be a polymath: Broaden your T of knowledge`
  - `Some niches are more difficult for AI`
    - `Cutting edge fields, novel idea generation, fields with less data online, ...`
  - `Focus on the human element`

Core concept

- The response to automation pressure is skill migration and systems-level capability, not passive resistance.
- The slide emphasizes augmentation, abstraction, and interdisciplinary breadth.
- It identifies domains where AI may remain weaker due to novelty or sparse data.

Detailed interpretation

- `Learn AI, don’t fear it` frames adaptation as knowledge acquisition rather than avoidance.
- The sub-bullet explicitly characterizes AI as an amplification tool rather than a mandatory replacement mechanism.
- `Fundamentals don’t fade` argues for enduring value in core principles despite tooling shifts.
- `Move “up the abstraction ladder”` points toward higher-level ownership of problem definition and solution architecture.
- `Thinking in Systems` highlights system components and integration points, which is directly relevant to architecture and multi-component engineering.
- `Be a polymath: Broaden your T of knowledge` advocates depth plus breadth.
- `Some niches are more difficult for AI` identifies edge domains such as cutting-edge work and novel ideation.
- `Focus on the human element` signals persistent value in human judgment, communication, and social dimensions, though the slide does not detail them further.

Technical implications / engineering relevance

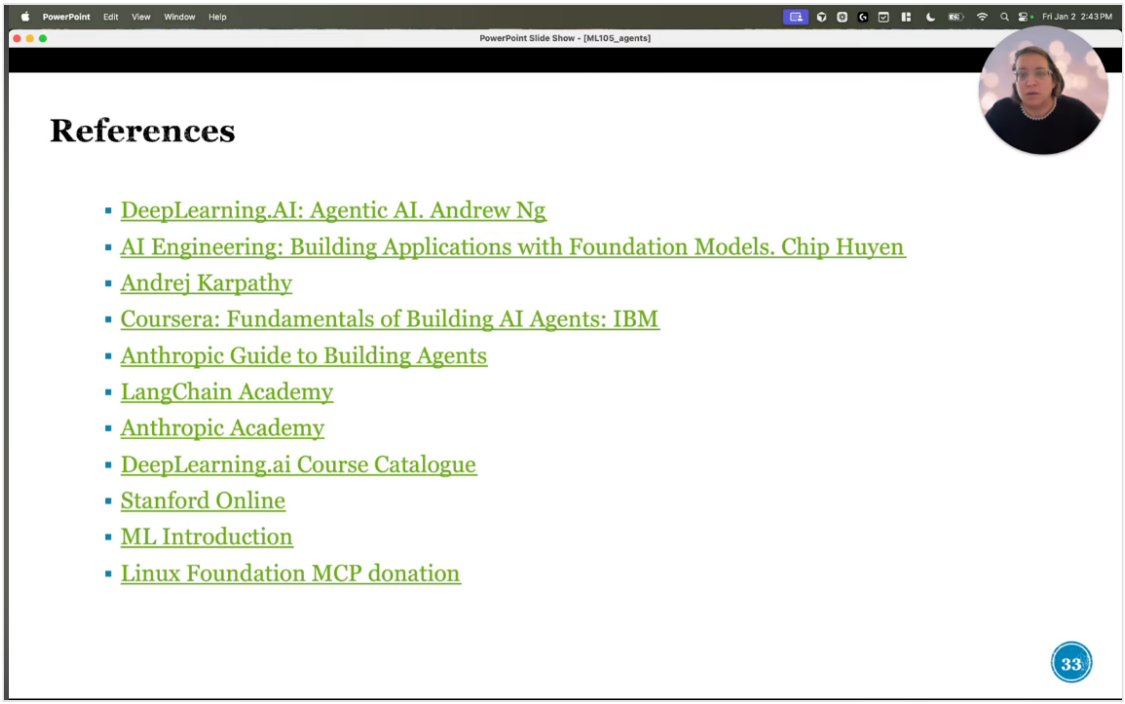
- Engineering roles may shift toward architecture, orchestration, integration, and outcome ownership.
- System design capability becomes more important as AI handles more narrow execution tasks.
- Cross-functional literacy supports better agent scoping, safer deployment, and stronger human-in-the-loop design.
- The mention of low-data and novel domains suggests caution when applying pretrained systems outside well-represented problem spaces.
- This slide is relevant to capability planning for teams building or operating agentic systems.

Key takeaways

- Adaptation favors abstraction, systems thinking, and broad technical literacy.
- AI augmentation does not eliminate the value of fundamentals.
- Novel, thin-data, and human-centric work may remain less tractable for AI.

Slide 33 - References

Image URI: `agentic_ai_assets/slide_33.png`



Main Point:
The slide provides a reading and learning-source list spanning agentic AI, AI engineering, courses, and interoperability-related material.

What the slide shows

- A title: `References`.
- A bullet list of visible references, including:
  - `DeepLearning.AI: Agentic AI. Andrew Ng`
  - `AI Engineering: Building Applications with Foundation Models. Chip Huyen`
  - `Andrej Karpathy`
  - `Coursera: Fundamentals of Building AI Agents: IBM`
  - `Anthropic Guide to Building Agents`
  - `LangChain Academy`
  - `Anthropic Academy`
  - `DeepLearning.ai Course Catalogue`
  - `Stanford Online`
  - `ML Introduction`
  - `Linux Foundation MCP donation`

Core concept

- The reference list anchors the slide deck in educational and ecosystem materials.
- The topics span practical agent construction, broader AI engineering, training resources, and at least one MCP-related ecosystem reference.
- The slide acts as provenance and further-study material rather than as an argument slide.

Detailed interpretation

- The references mix individuals, educational organizations, courses, and ecosystem resources.
- `Anthropic Guide to Building Agents` aligns with the deck's engineering orientation toward agent construction.

- `LangChain Academy`, `Anthropic Academy`, and `Coursera` entries indicate structured training routes.
- `Linux Foundation MCP donation` is the only explicit MCP-related reference on the slide and connects back to the earlier protocols discussion.
- Some entries are broad labels rather than full bibliographic citations; the slide appears to function as a practical reference list rather than a formal bibliography.
- No URLs are visible in the screenshot, so none are added here.

Technical implications / engineering relevance

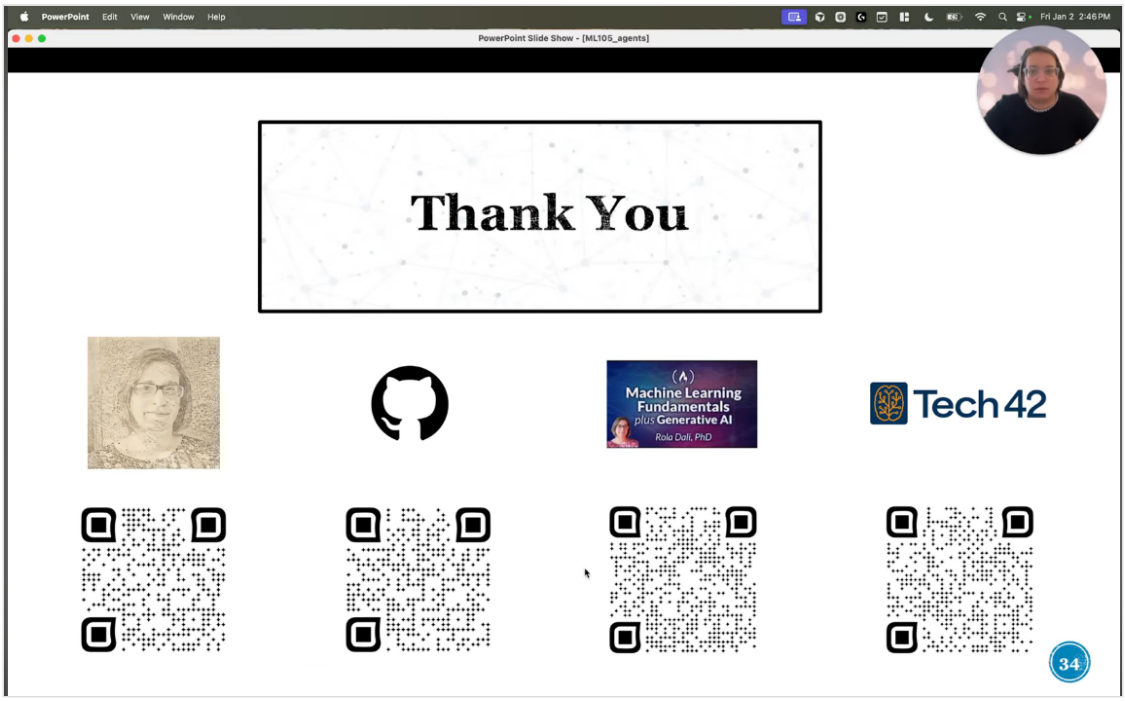
- The slide identifies a practical reading set for engineers building agentic systems.
- It also indicates that the deck draws from a mixed ecosystem of educational and implementation-oriented sources rather than a single framework.
- For handbook reuse, this slide can seed a curated resource appendix, though any full citation normalization would need source verification outside the image set.

Key takeaways

- The deck ends with a practical learning and reference set.
- The sources cover agents, AI engineering, education, and MCP-related ecosystem material.
- The visible slide content supports resource curation but not formal bibliographic expansion.

Slide 34 - Thank You

Image URI: `agentic_ai_assets/slide_34.png`



Main Point:
The closing slide provides a thank-you message and a set of QR-linked resources associated with the presenter and referenced platforms.

What the slide shows

- A centered `Thank You` message in a framed box.
- Four visual resource areas across the slide with corresponding QR codes beneath them.

- Visible logos/images including:
    - a portrait-style image
    - a GitHub logo
    - a `Machine Learning Fundamentals plus Generative AI` image
    - a `Tech42` logo
- Four QR codes aligned along the bottom half of the slide.

Core concept

- The slide is a contact/resource handoff slide rather than a content slide.
- It reinforces continuity beyond the presentation via scannable references.
- The resources appear to map to the presenter, code/resource platform, course material, and organization or brand references.

Detailed interpretation

- The slide does not provide visible destination text for the QR codes, so their exact targets cannot be confirmed from the image alone.
- The GitHub logo strongly suggests a code or repository-related destination.
- The course/book image suggests a learning resource destination.
- The Tech42 logo suggests an organizational or platform destination.
- Because QR destinations are unreadable from the screenshot without decoding, no exact URLs are claimed.

Technical implications / engineering relevance

- In handbook terms, this slide is best treated as a closing resource handoff artifact.
- It signals that practical follow-up materials may exist outside the slide deck.
- Any attempt to normalize the linked resources would require QR decoding or source access beyond the visible slide.

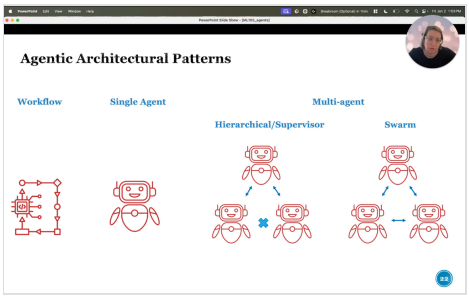
Key takeaways

- The closing slide is designed for resource follow-up rather than technical exposition.
- QR codes indicate external continuation points, but their destinations are not directly visible.
- No additional technical claims should be inferred from this slide alone.

Appendix - Image Reference by Chapter

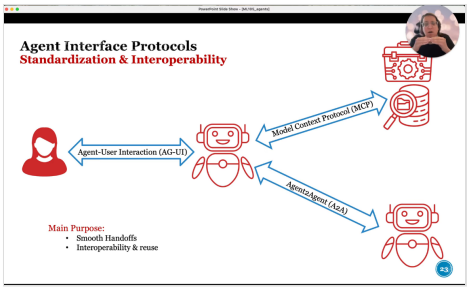
Appendix A - Slide 22 - Agentic Architectural Patterns

- Related chapter: `Slide 22 - Agentic Architectural Patterns`
- Image URI: `agentic_ai_assets/slide_22.png`
- Notes:
    - best available version used
    - no alternate capture from the provided set was required



Appendix A - Slide 23 - Agent Interface Protocols - Standardization & Interoperability

- Related chapter: `Slide 23 - Agent Interface Protocols - Standardization & Interoperability`
- Image URI: `agentic_ai_assets/slide_23.png`
- Notes:
    - best available version used for the body because it includes the `Main Purpose` bullets
    - alternate capture also exists: `Designing an Agent - Agent Interface Protokols -Building Agentic AI Workloads – Crash Course - YouTube.png`



Appendix A - Slide 24 - Evaluating Agents - Layers of the onion

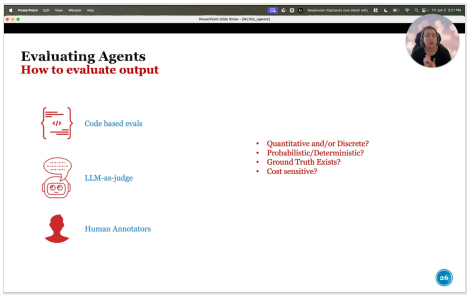
- Related chapter: `Slide 24 - Evaluating Agents - Layers of the onion`
- Image URI: `agentic_ai_assets/slide_24.png`
- Notes:
    - best available version used
    - no alternate capture from the provided set was required

Appendix A - Slide 25 - Evaluating Agents - What to evaluate

- Related chapter: `Slide 25 - Evaluating Agents - What to evaluate`
- Image URI: `agentic_ai_assets/slide_25.png`
- Notes:
    - best available version used
    - no alternate capture from the provided set was required

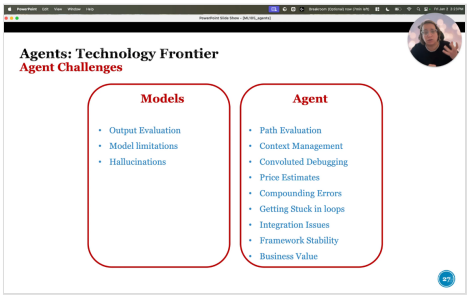
Appendix A - Slide 26 - Evaluating Agents - How to evaluate output

- Related chapter: `Slide 26 - Evaluating Agents - How to evaluate output`
- Image URI: `agentic_ai_assets/slide_26.png`
- Notes:
    - best available version used
    - no alternate capture from the provided set was required



Appendix A - Slide 27 - Agents: Technology Frontier - Agent Challenges

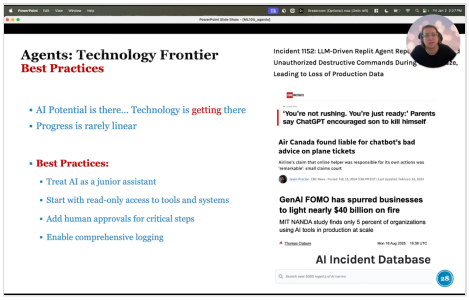
- Related chapter: `Slide 27 - Agents: Technology Frontier - Agent Challenges`
- Image URI: `agentic_ai_assets/slide_27.png`
- Notes:
    - best available version used
    - no alternate capture from the provided set was required



Appendix A - Slide 28 - Agents: Technology Frontier - Best Practices

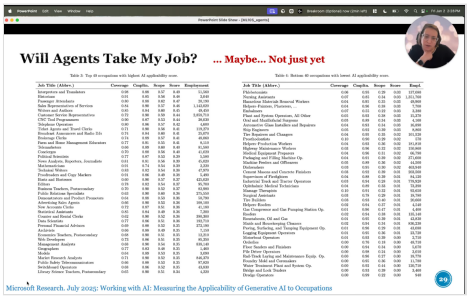
- Related chapter: `Slide 28 - Agents: Technology Frontier - Best Practices`
- Image URI: `agentic_ai_assets/slide_28.png`
- Notes:
    - best available slide screenshot used for body analysis
    - alternate related captures exist showing the same topic with different framing: `Designing an Agent - AI Incident Database - Incident Tracking - Building Agentic AI Workloads – Crash Course - YouTube.png` and `Designing an Agent -`

`Bestpractices - Bad behavour 1- Building Agentic AI Workloads – Crash Course - YouTube.png`



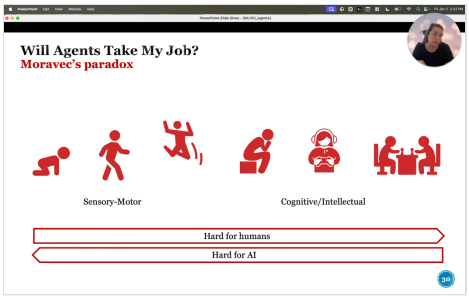
Appendix A - Slide 29 - Will Agents Take My Job? - ... Maybe... Not just yet

- Related chapter: `Slide 29 - Will Agents Take My Job? - ... Maybe... Not just yet`
- Image URI: `agentic_ai_assets/slide_29.png`
- Notes:
  - best available version used
  - no alternate capture from the provided set was required



Appendix A - Slide 30 - Will Agents Take My Job? - Moravec's paradox

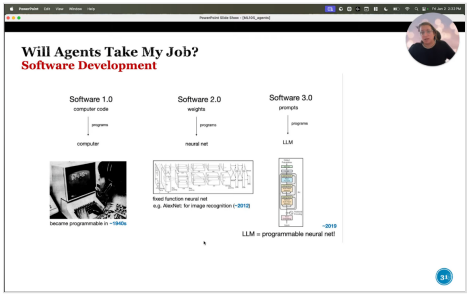
- Related chapter: `Slide 30 - Will Agents Take My Job? - Moravec's paradox`
- Image URI: `agentic_ai_assets/slide_30.png`
- Notes:
  - best available version used because it includes the lower difficulty arrows
  - alternate capture exists with tighter crop and without the lower arrow bars: `Design an Agent - Will Agent take my job - Building Agentic AI Workloads – Crash Course - YouTube.png`



Appendix A - Slide 31 - Will Agents Take My Job? - Software Development

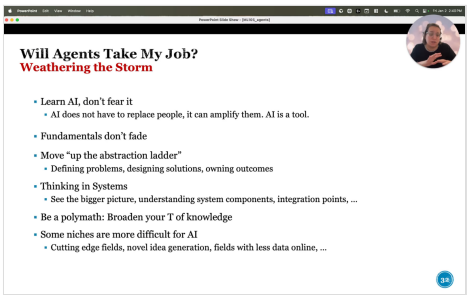
- Related chapter: `Slide 31 - Will Agents Take My Job? - Software Development`
- Image URI: `agentic_ai_assets/slide_31.png`

- Notes:
    - best available version used for primary analysis because the main three-column comparison is most readable
    - alternate captures exist with different side content or framing: `Design an Agent - Will agents take my job - Software Development - 2 Barcodes Backgrount - Building Agentic AI Workloads – Crash Course - YouTube.png` and `Design an Agent - Will agents take my job - Software Development 2 - Building Agentic AI Workloads – Crash Course - YouTube.png`



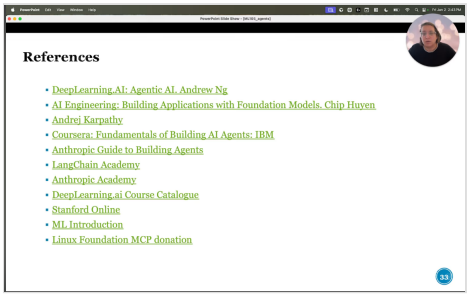
Appendix A - Slide 32 - Will Agents Take My Job? - Weathering the Storm

- Related chapter: `Slide 32 - Will Agents Take My Job? - Weathering the Storm`
- Image URI: `agentic_ai_assets/slide_32.png`
- Notes:
    - best available version used
    - alternate capture exists with slightly different framing: `Design an Agent - Will agents take my job - Weathering the storm - Building Agentic AI Workloads – Crash Course - YouTube..png`



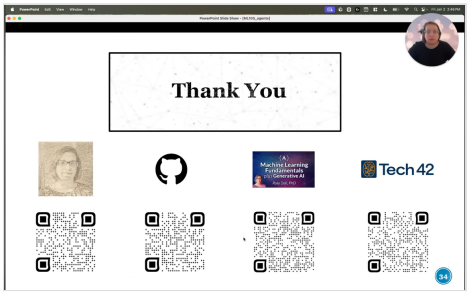
Appendix A - Slide 33 - References

- Related chapter: `Slide 33 - References`
- Image URI: `agentic_ai_assets/slide_33.png`
- Notes:
    - best available version used
    - no alternate capture from the provided set was required



Appendix A - Slide 34 - Thank You

- Related chapter: `Slide 34 - Thank You`
- Image URI: `agentic_ai_assets/slide_34.png`
- Notes:
    - best available version used
    - no alternate capture from the provided set was required



Final Verification

- Correct ordering:
    - confirmed from visible slide numbers 22 through 34
    - no missing numbered slide was invented
- Chapter completeness:
    - each ordered slide chapter contains image URI, main point, visible-content analysis, core concept, detailed interpretation, technical implications, and key takeaways
- Appendix completeness:
    - every body image appears again in the appendix
    - appendix entries include chapter references and duplicate/alternate notes where applicable
- Duplicate handling:
    - duplicate or alternate captures were not repeated as separate body chapters unless they represented materially different readable content, which they did not in this set
    - only one main chapter was retained per slide number
- Unsupported claims check:
    - no external technical detail was introduced unless grounded in visible labels or directly implied by the slide structure
    - uncertain items, such as unreadable QR destinations and unreadable fine-print table rows, were explicitly marked as uncertain